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The value of area-based analyses of donation patterns for recruitment strategies

A reading brief on where people donate, and why that should change how a blood service recruits them.

James AB, Josephson CD, Shaz BH, Schreiber GB, Hillyer CD, Roback JD. Transfusion. 2014;54(12):3051–3060. doi:10.1111/trf.12740.

Access to a donation site is not a background detail. In this Georgia study, the place a donor lives and the type of site they can reach are tied to whether they come back.

Full citation

[3] A. B. James, C. D. Josephson, B. H. Shaz, G. B. Schreiber, C. D. Hillyer, and J. D. Roback, "The value of area-based analyses of donation patterns for recruitment strategies," Transfusion, vol. 54, no. 12, pp. 3051–3060, 2014, doi: 10.1111/trf.12740.

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Record at a glance

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Journal	Transfusion (AABB), volume 54, issue 12, pages 3051–3060, December 2014
Design	Retrospective geographic analysis of whole-blood donations, with logistic regression
Place and years	Georgia, United States, donation years 2004–2008
Scale	402,692 donors and 1,147,442 whole-blood units
Blood service	The paper identifies American Red Cross Blood Services, Southern Region (ARCSR)
Main contrast	Metro Atlanta residents versus donors living outside metro Atlanta; fixed sites versus mobile sites

Question the paper asks

Most donor research describes people one at a time: age, sex, education, income, fear of needles, or a bad experience at the clinic. James and colleagues start from a different problem. A person may be willing to donate and still not donate often, because a collection site is hard to reach. The paper tests whether the geography of home and of the donation site is associated with how donors use fixed centers and mobile drives.

The authors treat the community, not only the individual, as a unit worth planning for. Zip code, city, and county carry different mixes of donors, incomes, and site access. A recruitment plan built only on a single "typical donor" will miss those local differences.

How the study was done

The population is people who gave whole blood in Georgia from 2004 through 2008. Each donor's home zip code, city, and county was matched to the address of the site where the unit was collected. Donors were split into two residence groups: metro Atlanta and nonmetro Atlanta.

Six location categories were defined from that match:

- Donation inside the same zip code as the donor's home, or in a different zip code.
- Donation inside the same city, or in a different city.
- Donation inside the same county, or in a different county.

Logistic regression compared donation patterns across those geographic units. The published abstract also reports a separate comparison that matters for operations: whether repeat donors used a fixed site or a mobile site, and whether that preference differed inside and outside metro Atlanta.

Read the p-values as evidence of association in this historical cohort. They do not, by themselves, prove that moving a drive would cause more donations. The design observes where units were already collected.

What the study found

The cohort is large enough that small operational differences are visible. Over five years the service recorded 402,692 whole-blood donors and 1,147,442 units. Just over half of the donors, 56.4 percent, lived in metro Atlanta. Three quarters of donors were white (75.0 percent), and 55.7 percent were female.

The result that should drive recruitment planning is the split by site type:

- Outside metro Atlanta, repeat donors were more likely to have donated at a fixed site ($p < 0.001$).
- Inside metro Atlanta, repeat donors were more likely to have donated at a mobile site than at a fixed site ($p < 0.001$).

The authors' conclusion is direct. Donation patterns differ by geography and by demography. Where a donor lives, and where the chair is set up, influences behavior. Area-based analysis is therefore a planning tool, not only a map.

What this means for recruitment

A fixed site and a mobile drive are not interchangeable channels. Outside the metro area, repeat donation lines up with a stable site. That fits a region where distance is long and a known center is easier to return to than a drive that moves. Inside metro Atlanta, repeat donors are reached more often through mobile collection. Workplace, campus, and community drives are the channel that already holds repeat donors in the city.

A practical reading for a blood service is:

- Map donors by home zip code, city, and county before choosing the next drive location. The paper's whole method is that match.

- Do not use one site mix for the whole state. The metro and nonmetro repeat-donor patterns point in opposite directions.
- Treat inconvenience as a structural barrier. The introduction places difficulty reaching a site alongside fear, anxiety, expected deferral, low awareness, and poor service.
- Build a community profile, not only a person profile. The textbook donor is often described as a middle-aged white man with a college education and a higher income. This cohort was majority female. Local composition will not match that textbook picture.
- Use the area analysis to decide where a fixed center is worth keeping and where a mobile calendar should be denser.

Limits to keep in view

- The data stop in 2008 and cover one state. Road networks, donor rules, and mobile-drive practice have changed since then.
- The abstract's headline statistics describe the cohort and the fixed-versus-mobile contrast. Odds ratios for each zip, city, and county comparison sit in the full paper and are not restated here.
- Residence was grouped as metro Atlanta or not. That is a coarse boundary. Two zip codes inside the same group can still differ in income, transit, and site density.
- The study records where donations happened. It cannot show the people who never donated because no site was near them.
- Whole blood is the product studied. Apheresis and platelet recruitment may follow a different geography.

How to use this reference

Cite this paper when you need evidence that recruitment should be planned by area, not only by a donor's personal traits. It supports three claims you can defend from the published record: donation-site access is associated with donation behavior; repeat donors in metro Atlanta and outside it do not use fixed and mobile sites the same way; and matching home geography to collection geography is a usable recruitment method.

Pair it with later work on travel time to blood banks, and with forecasting papers, when the argument moves from "where did donors already give?" to "where should the next campaign be scheduled?" James and colleagues answer the first question with Georgia data. They do not build a real-time prediction system.

One-paragraph summary you can reuse

James et al. (2014) matched home zip code, city, and county to collection-site addresses for 402,692 Georgia whole-blood donors (1,147,442 units, 2004–2008). Repeat donors outside metro Atlanta were more likely to use fixed sites, while repeat donors in metro Atlanta were more likely to use mobile sites (both $p < 0.001$). The paper's claim is that area-based analysis of these patterns is a practical tool for donor recruitment.

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